Trigonometry

Section 1: Trigonometric Identities & Simplifications

1 Simplify the trigonometric expression: $\frac{1}{1+\sin x} + \frac{1}{1-\sin x}$.

a. $2\sec^2 x$

b. $2\csc^2 x$

c. $2\cos^2 x$

d. 1

2 Which of the following is equal to $\tan x + \cot x$?

a. $\sec x \csc x$

b. $\sin x \cos x$

c. $\tan(2x)$

d. 1

3 If $\tan\theta = \frac{4}{3}$ and $\pi < \theta < \frac{3\pi}{2}$, what is $\sin(2\theta)$?

a. $\frac{24}{25}$

b. $-\frac{24}{25}$

c. $\frac{7}{25}$

d. $-\frac{7}{25}$

Section 2: Solving Trigonometric Equations

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4 Find all solutions to $2\cos^2 x + \cos x - 1 = 0$ in $[0, 2\pi)$.

a. $\frac{\pi}{3}, \frac{5\pi}{3}, \pi$

b. $\frac{\pi}{6}, \frac{11\pi}{6}, \pi$

c. $\frac{2\pi}{3}, \frac{4\pi}{3}$

d. $\frac{\pi}{4}, \frac{7\pi}{4}$

5 How many solutions does $\sin(3x) = \frac{1}{2}$ have in the interval $[0, 2\pi)$?

a. 2

b. 4

c. 6

d. 8

6 Solve $\tan^2 x = 3$ for $x \in [0, \pi]$.

a. $\frac{\pi}{3}, \frac{2\pi}{3}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}$ only

d. $\frac{\pi}{4}, \frac{3\pi}{4}$

Section 3: Laws of Sines & Cosines with Ambiguous Case

7 In $\triangle ABC$, $a = 12, b = 15,$ and angle $A = 30^\circ$. How many valid triangles can be formed?

a. 0

b. 1

c. 2

d. Infinitely many

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8 In $\triangle ABC$, $a = 5, b = 7,$ and $c = 8$. What is the measure of angle C?

a. 45°

b. 60°

c. 75°

d. 90°

9 What is the area of a parallelogram with side lengths 8 cm and 12 cm and an acute interior angle of 45°?

a. $48\sqrt{2}$ cm^2

b. 96 cm^2

c. 48 cm^2

d. $96\sqrt{2}$ cm^2

Section 4: Periodic Modeling & Phase Shifts

10 What is the phase shift of the sinusoidal wave $y = 3\sin(2x - \frac{\pi}{3}) + 4$?

a. $\frac{\pi}{6}$ to the right

b. $\frac{\pi}{3}$ to the right

c. $\frac{\pi}{6}$ to the left

d. $\frac{\pi}{2}$ to the right

11 What are the minimum and maximum values of the function $f(t) = 5 - 4\cos(3t)$?

a. Min $= 1$, Max $= 9$

b. Min $= -4$, Max $= 4$

c. Min $= 0$, Max $= 5$

d. Min $= -1$, Max $= 9$

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12 What is the frequency (cycles per unit of x) of the function $y = \cos(4\pi x)$?

a. 2

b. 4

c. $\frac{1}{2}$

d. $\frac{1}{4}$

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